

Worked Solutions

Discrete Mathematical Structures

Set B: Original | CSE Semester 3

Companion to dms-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Bottom of divisor lattice

Correct option: (C). 1 divides every element of this set.

2. Top of divisor lattice

Correct option: (B). Every element of the set divides 12.

3. Meet of divisors

Correct option: (A). The meet is the greatest common divisor, $\gcd(4, 6) = 2$.

4. Join of divisors

Correct option: (D). The join is the least common multiple, $\text{lcm}(4, 6) = 12$.

5. Complement existence

Correct option: (C). A complement c needs $\gcd(2, c) = 1$ and $\text{lcm}(2, c) = 12$. The only odd divisors, 1 and 3, give lcms 2 and 6; neither works.

6. Truth table and canonical forms

x	y	z	$\neg y$	$x \wedge \neg y$	F
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

Thus $F = \Sigma m(1, 3, 4, 5, 7) = \Pi M(0, 2, 6)$.

Canonical DNF: $F = (\neg x \wedge \neg y \wedge z) \vee (\neg x \wedge y \wedge z) \vee (x \wedge \neg y \wedge \neg z) \vee (x \wedge \neg y \wedge z) \vee (x \wedge y \wedge z)$.

Canonical CNF: $F = (x \vee y \vee z) \wedge (x \vee \neg y \vee z) \wedge (\neg x \vee \neg y \vee z)$.

7. Boolean identity proof

$$\begin{aligned}
 a \vee (\neg a \wedge b) &= (a \vee \neg a) \wedge (a \vee b) && \text{distributive law} \\
 &= 1 \wedge (a \vee b) && \text{complement law} \\
 &= a \vee b && \text{identity law.}
 \end{aligned}$$

8. Integer solutions by inclusion-exclusion

Set $u = x - 1$ and $v = y - 2$. Then $u + v + z = 9$, with $0 \leq u, v \leq 4$ and $z \geq 0$. The unrestricted count is $\binom{11}{2} = 55$. Violations $u \geq 5$ and $v \geq 5$ each leave a nonnegative sum of 4 and each contribute $\binom{6}{2} = 15$. Both cannot occur simultaneously because their sum would be at least 10. Therefore

$$55 - 15 - 15 = 25.$$

As a check, each of the five possible x values can pair with each of the five possible y values; even the maximum sum is $5 + 6 = 11 \leq 12$, and each pair fixes a nonnegative z .

9. 42nd lexicographic permutation

Use zero-based rank $42 - 1 = 41$. Factorial block choices are:

Available symbols	Block size	Quotient / remainder	Chosen
0,1,2,3,4	24	$41 = 1(24) + 17$	1
0,2,3,4	6	$17 = 2(6) + 5$	3
0,2,4	2	$5 = 2(2) + 1$	4
0,2	1	$1 = 1(1) + 0$	2
0	1	$0 = 0(1) + 0$	0

Choose the quotient-indexed symbol from the sorted remaining list at each step, with zero-based indices. The result is $\boxed{13420}$.

10. Arrangements of BALLOON

There are 7 letters, with L repeated twice and O repeated twice. The number of distinct arrangements is

$$\frac{7!}{2!2!} = 1260.$$

If the two Ls must be adjacent, treat LL as one object. The six objects are LL, B, A, O, O, N, so the count is

$$\frac{6!}{2!} = 360.$$

There is no additional factor for exchanging the identical Ls within their block.

11. Dijkstra on the undirected graph

Initialize A to 0 and every other tentative distance to infinity. All weights are nonnegative. After each settlement, relax each incident edge.

Settled	A	B	C	D	E
Initial	0	∞	∞	∞	∞
A	0	4	1	∞	∞
C	0	3	1	6	9
B	0	3	1	4	9
D	0	3	1	4	7
E	0	3	1	4	7

Destination	Shortest path	Distance
A	A	0
B	$A \rightarrow C \rightarrow B$	3
C	$A \rightarrow C$	1
D	$A \rightarrow C \rightarrow B \rightarrow D$	4
E	$A \rightarrow C \rightarrow B \rightarrow D \rightarrow E$	7

12. Connectivity proof

Suppose the graph is disconnected. If a component has s vertices, every vertex in it has degree at most $s - 1$. The minimum-degree assumption forces

$$s \geq \delta(G) + 1 \geq (n + 1)/2.$$

There are at least two components, so their combined size would be at least $n + 1$, contradicting the total of n vertices. Hence the graph is connected.

13. Degree and congruence

For a nonempty simple r -regular graph, its complement is $(n - 1 - r)$ -regular. Self-complementarity gives $r = n - 1 - r$, hence $r = (n - 1)/2$. Thus n is odd. By the handshake lemma, nr is even, so r must be even because n is odd. Writing $r = 2t$ gives $n = 4t + 1$, or $n \equiv 1 \pmod{4}$.